Лабораторная работа 6

Решение моделей в непрерывном и дискретном времени

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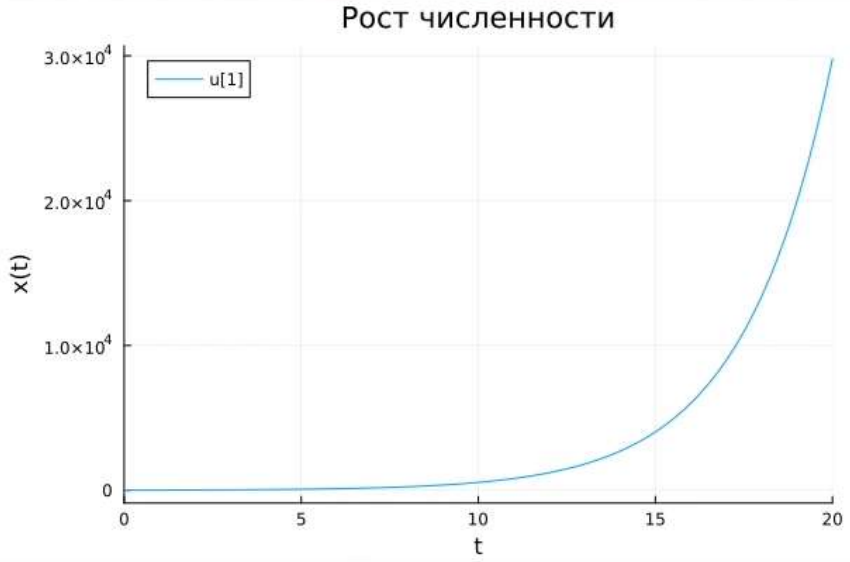
Цель работы

Основной целью работы является освоение специализированных пакетов для решения задач в непрерывном и дискретном времени.

Модель роста численности изолированной популяции (модель Мальтуса)

[8]:
```
a = 0.4
b = 0.6
c = 0.2

f(u, p, t) = a * u
x0 = 10
tspan = (0.0, 20.0)

prob = ODEProblem(f, x0, tspan)
sol = solve(prob)

plot(sol, xlabel="t", ylabel="x(t)", title = "Рост численности")
```

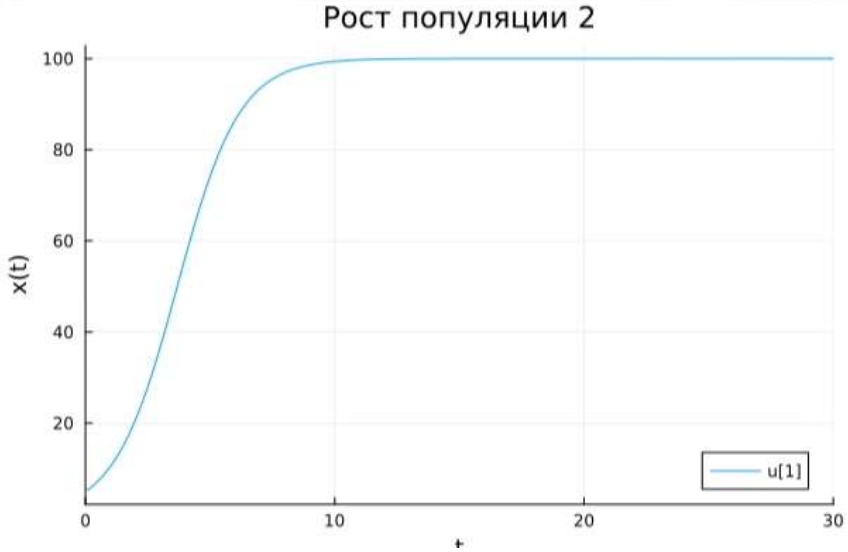
[8]:



Логистическая модель роста популяции

[11]:
```
r = 0.8
k = 100

function logistic!(du, u, p, t)
    du[1] = r*u[1]*(1-(u[1]/k))
end

u0 = [5.0]
tspan = (0.0, 30.0)

sol = solve(ODEProblem(logistic!,u0,tspan))

plot(sol, xlabel="t", ylabel="x(t)", title = "Рост популяции 2")
```

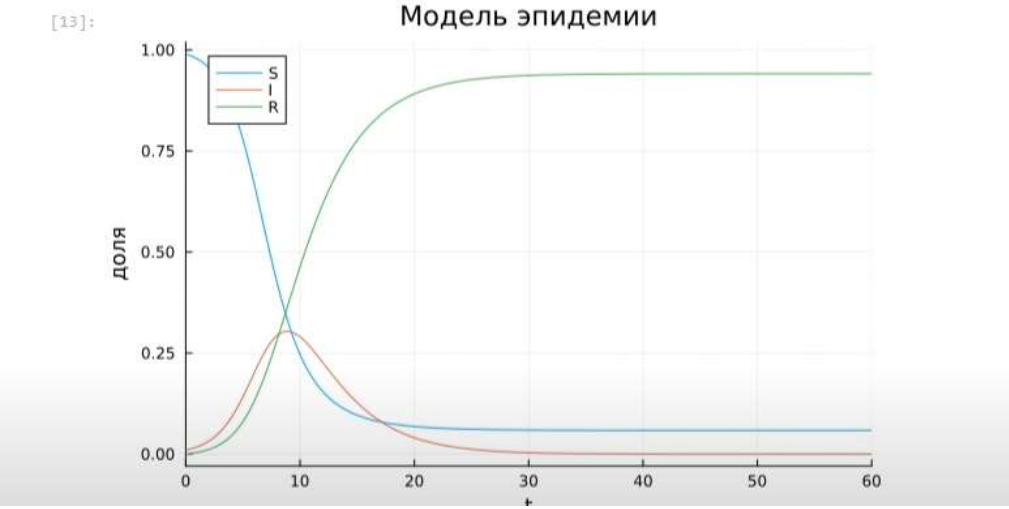
[11]:



Модель эпидемии Кермака–Маккендрика

[13]:
```
beta = 0.9
v = 0.3

function SIR!(du, u, p, t)
    s,i,r = u
    du[1] = -beta*i*s
    du[2] = beta*i*s - v*i
    du[3] = v*i
end

u0 = [0.99, 0.01, 0.0]
tspan = (0.0,60.0)

sol = solve(ODEProblem(SIR!,u0,tspan))

plot(sol, xlabel="t", ylabel="доля", title = "Модель эпидемии", label=["S" "I" "R"])
```

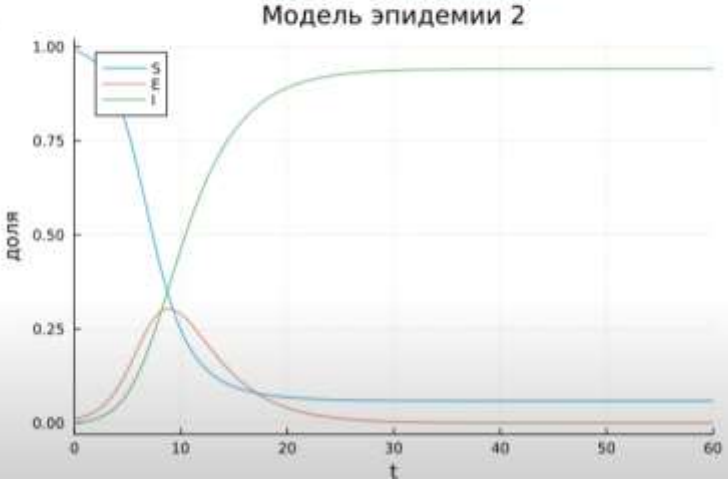
[13]:



Расширение модели эпидемии

```
[35]: beta = 0.9
delta = 0.4
gamma = 0.3
N = 1

function SEIR!(du, u, p, t)
    s,e,i,r = u
    du[1] = -(beta/N)*s*i
    du[2] = (beta/N)*s*i - delta*e
    du[3] = delta*e - gamma*i
    du[4] = gamma * i
end

u0 = [0.99, 0.0, 0.01, 0.0]
tspan = [0.0, 60.0]

sol2 = solve(ODEProblem(SEIR!,u0,tspan))
plot(sol, xlabel="t", ylabel="доля", title = "Модель эпидемии 2", label=["S" "E" "I" "R"])
```

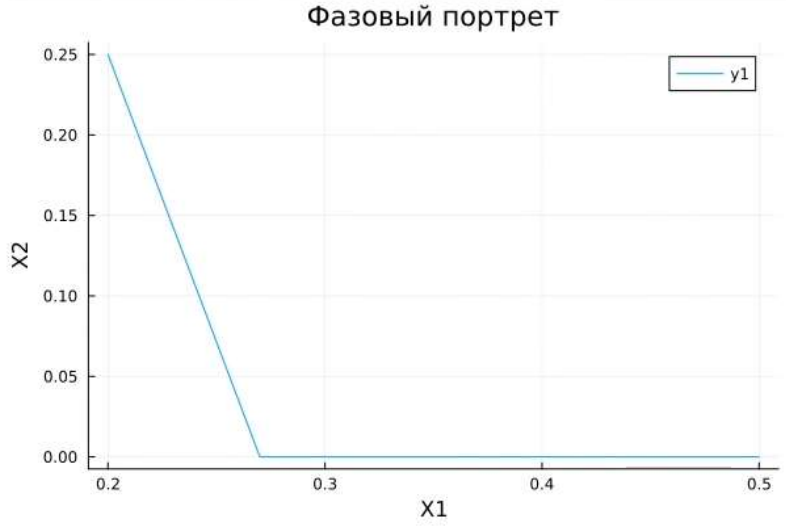
[35]:



Дискретная модель Лотки–Вольтерры

[20]:
```
a= 2
c = 1
d = 5

step(x1, x2) = (a*x1*(1-x1) - x1*x2, -c*x2 + d*x1*x2)
T = 200
x1 = zeros(T)
x2 = zeros(T)
x1[1], x2[1] = 0.2, 0.25

for t in 1:T-1
    x1[t+1], x2[t+1] = step(x1[t], x2[t])
end

plot(x1, x2, xlabel="X1", ylabel="X2", title="Фазовый портрет")
```

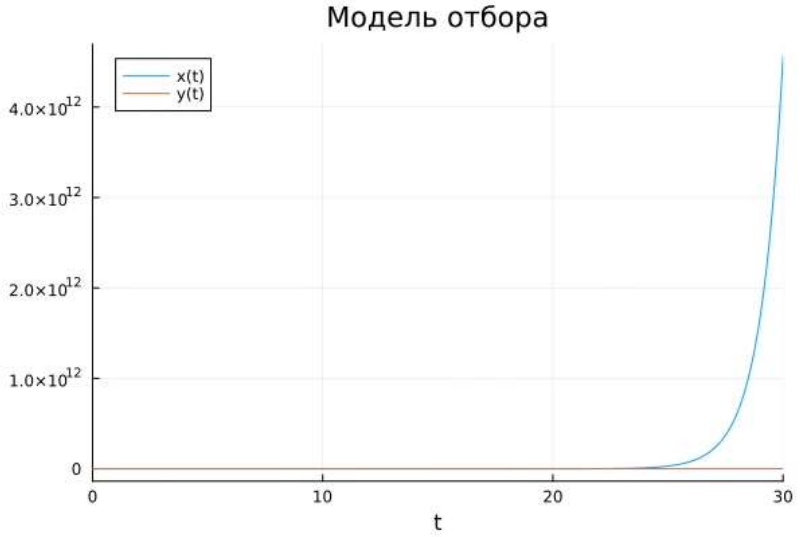
[20]:



Модель отбора

```
[21]: alpha = 1.0
beta = 0.6

function conc!(du, u, p, t)
    x,y = u
    du[1] = alpha*x - beta*x*y
    du[2] = alpha*y - beta*x*y
end

u0 = [1.2, 0.8]
tspan = (0.0, 30.0)
sol = solve(ODEProblem(conc!, u0, tspan))

plot(sol, label=["x(t)" "y(t)"], title="Модель отбора", xlabel = "t")
```

[21]:

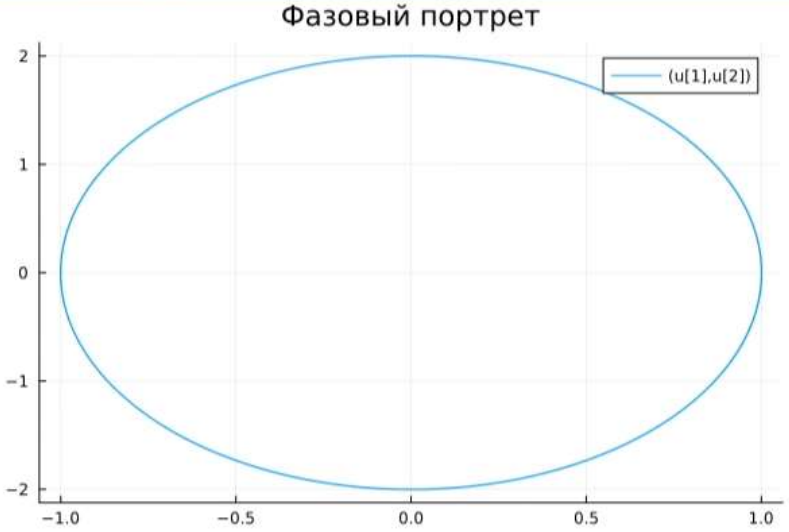


Модель консервативного гармонического осциллятора

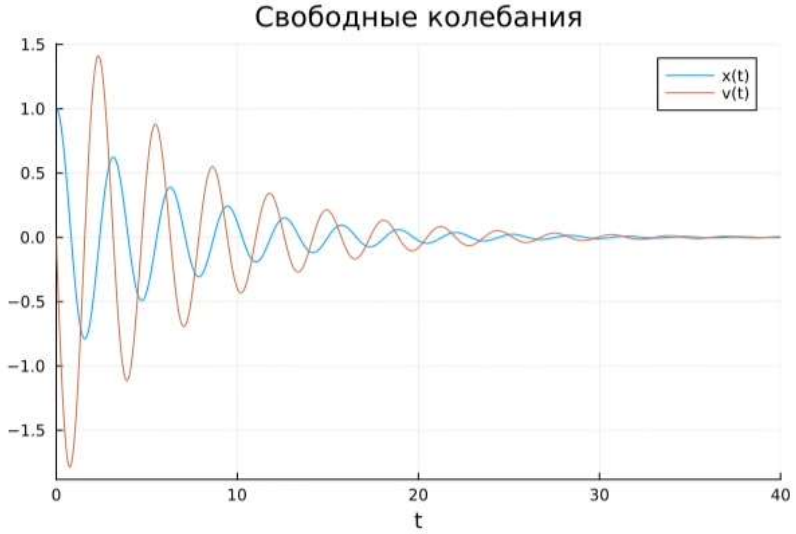
*[24]:

```
omega0 = 2.0

function osc!(du, u, p, t)
    x, v = u
    du[1] = v
    du[2] = -(omega0^2)*x
end

u0 = [1.0, 0.0]
tspan = [0.0, 20.0]
sol = solve(ODEProblem(osc!, u0, tspan))

plot(sol, label=["x(t)" "v(t)"], title="Гармонический осциллятор", xlabel="t")
```

[24]:



Модель свободных колебаний

[27]:
```
omega0 = 2.0
gamma = 0.15

function osc2!(du, u, p, t)
    x, v = u
    du[1] = v
    du[2] = -2gamma*v - (omega0^2)*x
end

u0 = [1.0, 0.0]
tspan = [0.0, 40.0]
sol = solve(ODEProblem(osc2!, u0, tspan))

plot(sol, label=["x(t)" "v(t)"], title="Свободные колебания", xlabel="t")
```

[27]: